



যাদবপুর বিশ্ববিদ্যালয়  
**JADAVPUR UNIVERSITY**

**ResTC-PQD: Residual TCN with Grad-CAM  
and SimCLR for Interpretable, Noise-Robust  
Single-Cycle PQD Classification**

Final Year Project Report

Submitted in partial fulfillment.

Bachelor of Engineering

in

Power Engineering

Submitted by

**Sabarna Saha**

(Roll No: 002211501176)

Under the supervision of

**DR. NILADRI CHAKRABARTY**

**Department of Power Engineering**

**JADAVPUR UNIVERSITY**

**2025–2026**

## **Vision and Mission**

### **Vision of the Institute**

To provide young minds an ambience and quality education in Engineering and Technology to contribute towards a better world.

### **Mission of the Institute**

- To nurture Engineering and Technological potential in undergraduate and post-graduate students at their highest standard.
- To take up technological challenges of the State, Nation, and beyond for ensuring social security and sustainable development.
- To provide infrastructure at par with international standards for quality training, research, and development.

### **Vision of the Department**

To emerge as a globally recognized department in imparting quality education to produce successful Power Engineers.

### **Mission of the Department**

- To provide the students with the state-of-the-art of enabling technologies related to energy and power engineering to meet the global challenges.
- To generate skilled human resources for sustainable development of the energy and allied sectors.
- To facilitate the students to choose a career in the industry, research and development, and entrepreneurship.

## **Program Outcomes & Objectives**

### **Program Educational Objectives (PEO)**

Graduates of Power Engineering Program shall:

- **PEO1:** Succeed in their career as globally employable power engineers and team leaders.
- **PEO2:** Pursue advanced education and research in energy, power, and allied interdisciplinary areas leading to lifelong learning successfully.
- **PEO3:** Have ethical values, social commitment, and leadership qualities towards application areas of electrical energy.

### **Program Specific Outcome (PSO)**

- **PSO1 Interdisciplinary Domain Exposure:** Interpret problems and apply enabling technologies to develop comprehensive solutions for the energy and power sectors.
- **PSO2 Economic and sustainable energy resources:** Assess and analyze energy resources and formulate optimized solutions for sustainable development.
- **PSO3 Safe and secured energy:** Recognize safety, control, and management aspects of new generation energy technology.

### **Program Outcomes (PO)**

1. **Engineering knowledge:** Apply knowledge of mathematics, science, and engineering fundamentals.
2. **Problem analysis:** Identify, formulate, and analyze complex engineering problems.
3. **Design & Development of Solutions:** Design solutions for complex engineering problems meeting specified needs.
4. **Investigation of Complex Problem:** Use research-based knowledge to provide valid conclusions.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques and IT tools.
6. **The Engineer and Society:** Assess societal, health, safety, and legal issues.
7. **Environment and Sustainability:** Understand the impact of professional engineering solutions in societal contexts.
8. **Ethics:** Apply ethical principles and commit to professional ethics.
9. **Individual and Team Work:** Function effectively as an individual or leader in diverse teams.
10. **Communication:** Communicate effectively on complex engineering activities.
11. **Project Management and Finance:** Apply engineering management principles.
12. **Life-long Learning:** Recognize the need for independent and life-long learning.

## Project Outcome

**Project Title:** ResTC-PQD: Residual TCN with Grad-CAM and SimCLR for Interpretable, Noise-Robust Single-Cycle PQD Classification

**Student Name:** Sabarna Saha (Roll: 002211501176)

### Updated Project Outcomes aligned with the four contributions:

1. Design and validate a 1D Residual Temporal Convolutional Network (TCN) for single-cycle classification of 17 power quality disturbance classes.
2. Develop a residual denoising autoencoder and noise-immunity profile to evaluate PQD classification across 15–50 dB SNR conditions.
3. Apply Grad-CAM explainability to verify that the classifier attends to physically meaningful temporal regions of PQD waveforms.
4. Investigate SimCLR self-supervised representation learning to quantify label efficiency when only limited annotated PQD data are available.

**Table 1:** Mapping of Project Outcomes to Program Outcomes (3: Strong, 2: Moderate, 1: Low)

| Project Outcome         | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| 1D Residual TCN         | 3   | 3   | 3   | 3   | 3   | –   | –   | –   | –   | 2    | –    | 2    |
| Noise Immunity + DAE    | 3   | 3   | 3   | 3   | 3   | 2   | 2   | –   | –   | 2    | –    | 2    |
| Grad-CAM Explainability | 2   | 3   | 2   | 3   | 3   | 2   | –   | 2   | –   | 3    | –    | 2    |
| SimCLR Label Efficiency | 3   | 3   | 3   | 3   | 3   | –   | –   | –   | –   | 3    | 2    | 3    |

## Certificate

This is to certify that the project report entitled “**ResTC-PQD: Residual TCN with Grad-CAM and SimCLR for Interpretable, Noise-Robust Single-Cycle PQD Classification**” has been submitted by **Sabarna Saha** (Roll No: **002211501176**) in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Power Engineering, Jadavpur University, during the academic year 2025–2026.

The work has been carried out under my supervision in the Department of Power Engineering, Jadavpur University. To the best of my knowledge, the matter embodied in this report has not been submitted elsewhere for the award of any degree or diploma.

---

Signature of Supervisor

Dr. Niladri Chakrabarty

---

Signature of Head of Department

Department of Power Engineering

## **Acknowledgement**

The Final Year Project is a major milestone in the academic journey of an engineering student. It provides an invaluable opportunity to apply theoretical knowledge and transform innovative ideas into meaningful contributions.

I express my heartfelt gratitude to my project guide, **Dr. Niladri Chakrabarty**, for granting me the freedom to choose a project topic aligned with my interest. His continuous guidance, constructive feedback, and encouragement throughout the development of this project have been instrumental in its success. The four new contributions developed in this final version—the 1D Residual TCN, the noise-immunity profile with residual DAE, Grad-CAM explainability, and SimCLR label-efficiency analysis—were shaped by his guidance on technical depth, engineering relevance, and rigorous validation.

I also acknowledge the support provided by the Department of Power Engineering for the academic environment that enabled me to carry out this project smoothly. Lastly, I extend my thanks to my family and friends for their constant motivation.

**Sabarna Saha**

(Roll: 002211501176)

## Abstract

This project presents a single-cycle Power Quality Disturbance (PQD) classification framework for 17 disturbance classes from the XPQRS dataset. Each waveform contains 999 samples representing one 50 Hz cycle (20 ms), corresponding to an effective sampling rate of 49,950 Hz. The final evaluation uses a 70/15/15 stratified split yielding 1190 train, 255 validation, and 255 test samples. The work introduces four contributions. First, it replaces transform-heavy CNN–LSTM processing with a 1D Residual Temporal Convolutional Network (TCN) that learns directly from normalised waveforms using seven dilated residual blocks with dilation rates  $d = \{1, 2, 4, 8, 16, 32, 64\}$ . Second, it builds a residual denoising autoencoder (DAE) trained on a discrete set of SNR values  $\{20, 25, 30, 35, 40, 45, 50\}$  dB and evaluates robustness over 15–50 dB SNR. Third, it adapts Grad-CAM to one-dimensional temporal activations to produce saliency maps showing which waveform regions drive classification decisions. Fourth, it applies inductive SimCLR self-supervised learning with four domain-informed augmentations to study label efficiency under limited annotation. The final TCN achieves 99.61% test accuracy on 255 evaluation samples (254/255 correct), confirmed by the attached confusion matrix. The noise profile remains flat across all tested SNR levels, with no measurable DAE gain under AWGN. Grad-CAM saliency maps are consistent with physical PQD signatures, while inductive SimCLR improves scratch training most clearly at the 20–50% label fractions.

# Contents

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| <b>Vision and Mission</b>                                             | <b>1</b>  |
| <b>Program Outcomes &amp; Objectives</b>                              | <b>2</b>  |
| <b>Project Outcome</b>                                                | <b>3</b>  |
| <b>Certificate</b>                                                    | <b>4</b>  |
| <b>Acknowledgement</b>                                                | <b>5</b>  |
| <b>Abstract</b>                                                       | <b>6</b>  |
| <b>1 Introduction</b>                                                 | <b>9</b>  |
| 1.1 Background . . . . .                                              | 9         |
| 1.2 Need for Single-Cycle Detection . . . . .                         | 9         |
| 1.3 Limitations of Existing Methods . . . . .                         | 9         |
| 1.4 Proposed Contributions . . . . .                                  | 10        |
| 1.5 Chapter Organisation . . . . .                                    | 10        |
| <b>2 Literature Review</b>                                            | <b>11</b> |
| 2.1 Classical Signal Processing Approaches . . . . .                  | 11        |
| 2.2 Machine Learning with Handcrafted Features . . . . .              | 11        |
| 2.3 Deep Learning for PQD Classification . . . . .                    | 11        |
| 2.4 Temporal Convolutional Networks . . . . .                         | 12        |
| 2.5 Explainability in Deep Learning Models . . . . .                  | 12        |
| 2.6 Self-Supervised and Contrastive Representation Learning . . . . . | 12        |
| 2.7 Research Gaps Addressed by This Work . . . . .                    | 13        |
| <b>3 Dataset and Preprocessing</b>                                    | <b>14</b> |
| 3.1 XPQRS Dataset . . . . .                                           | 14        |
| 3.2 Class Labels and Physical Meaning . . . . .                       | 14        |
| 3.3 Preprocessing Pipeline . . . . .                                  | 15        |



|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| <b>4</b> | <b>Methodology</b>                                                | <b>16</b> |
| 4.1      | System Overview . . . . .                                         | 16        |
| 4.2      | Residual Denoising Autoencoder (DAE) . . . . .                    | 16        |
| 4.2.1    | Architecture and Training . . . . .                               | 16        |
| 4.3      | Stage 1: Derivative Features and SVM (Proposed Cascade) . . . . . | 17        |
| 4.4      | Contribution 1: 1D Residual TCN (Stage 2) . . . . .               | 17        |
| 4.4.1    | Architecture . . . . .                                            | 17        |
| 4.5      | Contribution 2: Noise Immunity Profile . . . . .                  | 19        |
| 4.6      | Contribution 3: Grad-CAM Explainability . . . . .                 | 19        |
| 4.7      | Contribution 4: SimCLR Self-Supervised Learning . . . . .         | 20        |
| <b>5</b> | <b>Results and Discussion</b>                                     | <b>24</b> |
| 5.1      | TCN Classification Performance . . . . .                          | 24        |
| 5.2      | Noise Robustness Analysis . . . . .                               | 25        |
| 5.3      | Explainability Analysis . . . . .                                 | 26        |
| 5.4      | SimCLR Label Efficiency Discussion . . . . .                      | 26        |
| 5.5      | Comparison with Prior Work . . . . .                              | 27        |
| <b>6</b> | <b>Applications</b>                                               | <b>28</b> |
| 6.1      | Smart Energy Meters . . . . .                                     | 28        |
| 6.2      | Digital Protection Relays . . . . .                               | 28        |
| 6.3      | Industrial Load Monitoring . . . . .                              | 28        |
| 6.4      | Renewable Energy Integration . . . . .                            | 28        |
| 6.5      | Preventive Maintenance . . . . .                                  | 28        |
| <b>7</b> | <b>Conclusion and Future Scope</b>                                | <b>29</b> |
| 7.1      | Conclusion . . . . .                                              | 29        |
| 7.2      | Future Scope . . . . .                                            | 29        |
|          | <b>References</b>                                                 | <b>31</b> |

# Chapter 1

## Introduction

### 1.1 Background

Modern electric grids operate with increasing penetration of power electronic converters, renewable energy interfaces, adjustable-speed drives, electric vehicle chargers, switched-mode power supplies, and non-linear industrial loads. These devices improve controllability and efficiency, but they also inject disturbances that degrade voltage quality. Power Quality Disturbances (PQDs) such as sag, swell, interruption, flicker, harmonics, notches, transients, oscillatory transients, and hybrid combinations can trip protection devices, corrupt data-centre supply, damage semiconductor fabrication equipment, and reduce the lifetime of grid-connected converters.

A practical PQD classifier must identify not only isolated events but also combined events. The XPQRS benchmark used in this project contains 17 classes, including hybrid classes such as *Sag\_with\_Harmonics*, *Flicker\_with\_Swell*, and *Swell\_with\_Oscillatory\_Transient*. These hybrid signatures challenge simple threshold logic because they combine amplitude deviation, spectral distortion, and short-duration time-localised behaviour in the same single-cycle record.

### 1.2 Need for Single-Cycle Detection

Protection relays and smart meters do not have the luxury of waiting for long observation windows. A 50 Hz power system completes one cycle in 20 ms. If a classifier can make a reliable decision from one cycle, then it can support relay logic, alarm generation, event tagging, and local mitigation with low latency. Multi-cycle methods increase evidence but also delay action. For sensitive loads, the difference between one cycle and several cycles can determine whether an event remains a logged disturbance or becomes equipment malfunction.

The single-cycle requirement also supports embedded deployment. A smart meter, relay, or edge monitoring unit can buffer one waveform segment, normalise it, run the classifier, and transmit only the event label and confidence score. This reduces memory usage, communication bandwidth, and cloud dependence.

### 1.3 Limitations of Existing Methods

Classical methods based on Wavelet Transform (WT), S-Transform, Empirical Mode Decomposition, and handcrafted statistics provide strong signal-processing insight, but they often require careful transform selection, several cycles of data, or computationally expensive time–frequency maps. CNN–LSTM pipelines improve feature learning, but many such approaches depend on Continuous Wavelet Transform (CWT) preprocessing before classification. That transform increases latency and memory traffic. LSTM layers further process sequences recurrently, which limits parallelism during inference.

Several published methods report clean-data accuracy but do not provide a systematic SNR sweep. Without noise testing, a model may perform well on curated synthetic waveforms but fail under mea-

surement noise, switching noise, or sensor quantisation. Many methods also report only final accuracy, leaving the decision process opaque. For engineering deployment, explainability matters because relay engineers need confidence that a model uses physically meaningful evidence. Finally, most supervised deep networks assume enough labelled samples, whereas actual utility waveforms are overwhelmingly unlabelled.

## 1.4 Proposed Contributions

This report presents the final implementation as four engineering contributions:

- **Contribution 1: 1D Residual TCN.** A seven-block dilated residual TCN replaces transform-heavy CNN–LSTM processing and classifies single-cycle waveforms directly, achieving 99.61% test accuracy on a strictly separate test set.
- **Contribution 2: Noise Immunity Profile.** A residual DAE and SNR sweep quantify robustness from 15 dB to 50 dB, instead of reporting only clean-data accuracy; the final AWGN sweep shows no measurable DAE gain over the TCN-only path.
- **Contribution 3: Grad-CAM Explainability.** A 1D Grad-CAM adaptation visualises temporal saliency and connects model decisions to PQD physics.
- **Contribution 4: SimCLR Label Efficiency.** An inductive self-supervised contrastive module tests whether useful representations can be learned from unlabelled waveform records, with the clearest gains over scratch training at 20–50% label fractions.

## 1.5 Chapter Organisation

Chapter 2 reviews classical, machine-learning, deep-learning, and self-supervised literature. Chapter 3 presents the XPQRS dataset and preprocessing pipeline. Chapter 4 details the DAE, SVM cascade concept, 1D Residual TCN, noise profile, Grad-CAM, and SimCLR modules. Chapter 5 discusses classification accuracy, noise robustness, explainability, label efficiency, and comparison with prior work. Chapter 6 presents applications in smart meters, protection relays, industrial monitoring, renewable integration, and preventive maintenance. Chapter 7 concludes the work and identifies future scope.

## Chapter 2

### Literature Review

#### 2.1 Classical Signal Processing Approaches

Early PQD detection relied on time–frequency transforms that expose both spectral and temporal information. The *S-Transform*, introduced by Stockwell et al. and applied to power quality by Dash et al. [4], combines a frequency-dependent Gaussian window with Fourier analysis to achieve time–frequency localisation superior to short-time Fourier analysis. It correctly identifies dominant harmonic and sag signatures, but its computational complexity grows quadratically with signal length, making it unsuitable for real-time single-cycle decisions on resource-constrained hardware.

Wavelet Transform (WT) methods offer a multi-resolution decomposition that captures transient events well [15]. Several authors use discrete wavelet coefficients at multiple scales as classifier inputs. However, the results depend strongly on the choice of mother wavelet, and overlapping hybrid disturbance classes can produce similar wavelet coefficient distributions. Empirical Mode Decomposition (EMD) and Hilbert–Huang Transform (HHT) adapt to non-stationary signals without a fixed basis, but mode mixing and sensitivity to noise limit their practical use in automated PQD pipelines.

#### 2.2 Machine Learning with Handcrafted Features

Support Vector Machines became a standard benchmark for PQD classification once reliable feature extraction methods were established [3]. A compact descriptor vector, typically containing RMS magnitude, total harmonic distortion, spectral entropy, and wavelet-band energies, feeds an SVM decision boundary. The resulting classifiers are interpretable and fast, but they require careful domain-knowledge-driven feature selection and are limited to patterns captured in the predefined descriptor.

The XPQRS framework of Khan et al. [1] is the closest baseline to this project. It demonstrated that finite-difference derivatives of the voltage waveform amplify disturbance signatures that are invisible in the raw signal. Combined with log energy, Shannon energy, and Hjorth mobility computed on derivative sequences, a quadratic SVM achieved 96.5% accuracy on 17-class XPQRS data using only a single 20 ms cycle. This project directly adopts the XPQRS feature philosophy in Stage 1 and builds on it with a deep learning Stage 2.

#### 2.3 Deep Learning for PQD Classification

Convolutional Neural Networks and LSTM-based architectures progressively replaced handcrafted features in PQD classification from 2018 onwards. Sahoo et al. [6] proposed a deep-learning architecture that learns PQD representations directly, achieving high accuracy on benchmark datasets. CNN models applied to raw waveforms learn local temporal patterns that correspond roughly to transform features, but they can overfit on small datasets without careful regularisation.

Salles and Ribeiro [14] combined two-dimensional wavelet scalograms with deep convolutional networks, demonstrating that treating a 1-D waveform as a 2-D time–frequency image can improve feature

richness. The limitation is the additional computational burden of the wavelet transform, which adds significant latency for a single-cycle decision. Xiao and Li [11] integrated multiple feature extraction modules into a unified deep pipeline that processes raw and transformed inputs jointly, achieving state-of-the-art accuracy at the cost of increased model complexity.

Chiam et al. [12] combined wavelet preprocessing with an LSTM network and an attention mechanism. The attention weights provided a limited form of explainability by highlighting high-energy wavelet subbands. However, LSTM-based inference processes time steps sequentially and therefore cannot exploit modern parallel computation as fully as convolutional alternatives. A comprehensive survey by Chawda et al. [10] covering deep learning methods for PQD in utility grids with renewable penetration explicitly identified interpretability and noise robustness as open challenges.

## **2.4 Temporal Convolutional Networks**

Bai et al. [9] provided the canonical empirical evaluation comparing dilated causal TCNs with LSTMs and GRUs across diverse sequence benchmarks. Their results showed that a carefully tuned TCN consistently matched or exceeded LSTM performance while enabling fully parallel training and inference. For a fixed-length observation window such as a single power cycle, causal masking is unnecessary because the full segment is available at inference time. The present work therefore uses non-causal symmetric padding, which allows each filter position to use bilateral context and yields a more symmetric receptive field at equal depth. Dilated convolutions with exponentially growing dilation rates allow a receptive field that grows logarithmically in the number of parameters, enabling efficient coverage of the 999-sample window without excessive depth or parameter count.

## **2.5 Explainability in Deep Learning Models**

Selvaraju et al. [8] introduced Gradient-weighted Class Activation Mapping (Grad-CAM) for spatial image explanation. The method computes the gradient of a class score with respect to a convolutional feature map and pools these gradients to weight activations. The resulting heat map highlights input regions relevant to the decision. In one-dimensional waveform settings, the spatial interpretation is replaced by a temporal one: the heat map localises the milliseconds of the waveform most responsible for the classification outcome.

Very few published PQD classifiers include any form of gradient-based explainability. Chawda et al. [10] identified interpretability as an open challenge for deep learning in power quality monitoring, noting that grid operators require evidence-based decisions for safety-critical relay tripping. The Grad-CAM component of this project directly addresses this gap by producing per-sample temporal saliency maps aligned with domain-expected disturbance signatures.

## **2.6 Self-Supervised and Contrastive Representation Learning**

Chen et al. [7] proposed SimCLR, a simple framework for visual contrastive learning: augment each sample twice, maximize agreement between the two views via the NT-Xent loss, and use the encoder representation for downstream classification tasks. The approach benefits from large batch sizes and datasets but has since been adapted for audio, medical imaging, and time-series domains.

For general time-series, Eldele et al. [13] extended contrastive learning with temporal and contextual contrasting modules, demonstrating improved downstream classification under limited labels across several benchmark datasets. In the power quality context, self-supervised approaches have not yet been widely studied, even though grid operators routinely collect terabyte-scale waveform archives with sparse annotation. The SimCLR component of this project addresses this gap by testing whether four physically motivated augmentations can encode PQD-class invariances in a label-free encoder trained on only 1700 waveforms.

## 2.7 Research Gaps Addressed by This Work

Table 2.1 consolidates the key attributes and limitations of selected prior works and identifies the specific contributions of this project.

**Table 2.1:** Literature comparison identifying gaps addressed by this work. Accuracy values are those reported in the respective publications on their own datasets; they are not directly comparable across different benchmarks.

| Method                | Input               | Classes   | Reported accuracy                       | Noise tested         | XAI             | Single-cycle | Primary limitation                     |
|-----------------------|---------------------|-----------|-----------------------------------------|----------------------|-----------------|--------------|----------------------------------------|
| Dash et al. [4]       | S-Transform         | PQ events | Event analysis                          | No                   | No              | No           | High compute; limited classes.         |
| Khan et al. [1]       | Derivative features | 17        | 96.5%                                   | Limited              | No              | Yes          | Handcrafted only; no deep path.        |
| Sahoo et al. [6]      | Learned DL features | Multi     | Not independently verified <sup>†</sup> | No                   | No              | No           | Clean-data only; black-box.            |
| Salles & Ribeiro [14] | 2-D wavelet         | Multi     | Reported >99% on own dataset            | No                   | No              | No           | CWT preprocessing cost.                |
| Chiam et al. [12]     | Wavelet + LSTM      | Multi     | ≈98.5% on own dataset                   | No                   | Attention       | No           | Sequential LSTM; transform dependency. |
| Eldele et al. [13]    | Raw time-series     | Multi     | Repr. study                             | N/A                  | No              | N/A          | Not validated on PQD constraints.      |
| <b>This work</b>      | Raw waveform        | <b>17</b> | <b>99.61% on XPQRS</b>                  | <b>Yes, 15–50 dB</b> | <b>Grad-CAM</b> | <b>Yes</b>   | Synthetic dataset only.                |

<sup>†</sup>Exact accuracy from Sahoo et al. 2020 was not independently verified.

The selected comparison indicates that no listed entry combines all four of: direct 1-D waveform inference without transform preprocessing, systematic SNR profiling, Grad-CAM temporal saliency, and SimCLR label-efficiency analysis.

## Chapter 3

### Dataset and Preprocessing

#### 3.1 XPQRS Dataset

The implementation uses the XPQRS dataset distributed via the SEED Power Quality Disturbance dataset on Kaggle. Class-wise CSV files are loaded, and the resulting waveform matrix contains 1700 samples across 17 classes. Each instance contains **999 samples representing one 50 Hz cycle (20 ms)**, corresponding to an effective sampling rate of **49,950 Hz** (49.95 kHz). The final run uses a 70/15/15 stratified split with 1190 training samples, 255 validation samples, and 255 test samples. Validation is used for all callbacks and early stopping; the test set is touched exactly once at the end. Each class contributes exactly 15 samples to the test partition.

*Note on the original XPQRS paper:* Khan et al. [1] describe their synthetic generator as using 5 kHz, which would yield 100 samples per 20 ms cycle. The Kaggle dataset used here has 999 samples per instance, consistent with a  $\approx 50$  kHz effective rate. All reported results are based on the Kaggle dataset as distributed.

#### 3.2 Class Labels and Physical Meaning

The 17 class names are kept exactly as they appear in the final confusion matrix.

**Table 3.1:** XPQRS class list and physical descriptions.

| No. | Class name                       | Physical description                                                                           |
|-----|----------------------------------|------------------------------------------------------------------------------------------------|
| 1   | Flicker                          | Low-frequency voltage modulation causing visible light flicker and load sensitivity.           |
| 2   | Flicker_with_Sag                 | Voltage magnitude depression combined with slow modulation.                                    |
| 3   | Flicker_with_Swell               | Voltage magnitude rise combined with slow modulation.                                          |
| 4   | Harmonics                        | Non-sinusoidal waveform caused by integer-multiple frequency components from non-linear loads. |
| 5   | Harmonics_with_Sag               | Harmonic distortion combined with voltage sag.                                                 |
| 6   | Harmonics_with_Swell             | Harmonic distortion combined with voltage swell.                                               |
| 7   | Interruption                     | Severe voltage reduction approaching zero for a short duration.                                |
| 8   | Notch                            | Periodic narrow voltage cuts commonly associated with converter commutation.                   |
| 9   | Oscillatory_Transient            | Short-duration high-frequency oscillation superimposed on the voltage waveform.                |
| 10  | Pure_Sinusoidal                  | Normal reference waveform without disturbance.                                                 |
| 11  | Sag                              | RMS voltage decrease below nominal level for part of the cycle.                                |
| 12  | Sag_with_Harmonics               | Sag event with simultaneous harmonic distortion.                                               |
| 13  | Sag_with_Oscillatory_Transient   | Sag combined with a localised oscillatory transient.                                           |
| 14  | Swell                            | RMS voltage increase above nominal level for part of the cycle.                                |
| 15  | Swell_with_Harmonics             | Swell event with simultaneous harmonic distortion.                                             |
| 16  | Swell_with_Oscillatory_Transient | Swell combined with a localised oscillatory transient.                                         |
| 17  | Transient                        | Short-duration impulsive deviation caused by switching, faults, or lightning-like events.      |

### 3.3 Preprocessing Pipeline

For each waveform  $x$ , the notebook applies per-sample normalisation:

$$\tilde{x} = \frac{x - \mu_x}{\sigma_x + 10^{-8}}, \quad (3.1)$$

where  $\mu_x$  and  $\sigma_x$  are computed over the 999 samples of the same instance. This operation removes waveform-specific DC shift and scale differences while preserving temporal shape. The normalised waveform is reshaped to (999, 1) for Conv1D processing.

A normalisation ablation was performed using a 20-epoch mini-TCN on the validation set. The results show that per-instance z-score normalisation is not the best strategy overall: global z-score normalisation achieves the highest validation accuracy, while per-instance z-score removes absolute amplitude information that is physically important for sag and swell recognition.

**Table 3.2:** Normalisation ablation using a 20-epoch mini-TCN on the validation set.

| Normalisation strategy | Val accuracy  | Sag recall | Swell recall |
|------------------------|---------------|------------|--------------|
| Per-instance z-score   | 70.20%        | 80.0%      | 86.7%        |
| Per-instance min–max   | 63.53%        | 33.3%      | 40.0%        |
| Global z-score         | <b>76.47%</b> | 66.7%      | 86.7%        |

Global z-score normalisation is therefore the best strategy by validation accuracy (76.47%), compared with 70.20% for per-instance z-score. Per-instance z-score is retained for the main model for consistency with prior results, but the ablation corroborates the reviewer concern that removing per-instance amplitude information can hurt amplitude-defined classes. Full re-training under global normalisation is identified as future work.

Finite-difference derivatives are computed as:

$$\Delta^0 x[t] = x[t], \quad \Delta^n x[t] = \Delta^{n-1} x[t] - \Delta^{n-1} x[t-1], \quad n = 1, 2, 3, 4, \quad (3.2)$$

where boundary samples ( $t < n$ ) are zero-padded on the left so that all derivative vectors retain the original length of 999.

The stratified 70/15/15 split preserves class balance. Since 1700 samples are distributed uniformly over 17 classes, each class contains 100 examples. The test set therefore contains 15 examples per class, which enables a clean per-class confusion matrix interpretation where every off-diagonal entry is directly interpretable as a misclassification rate.

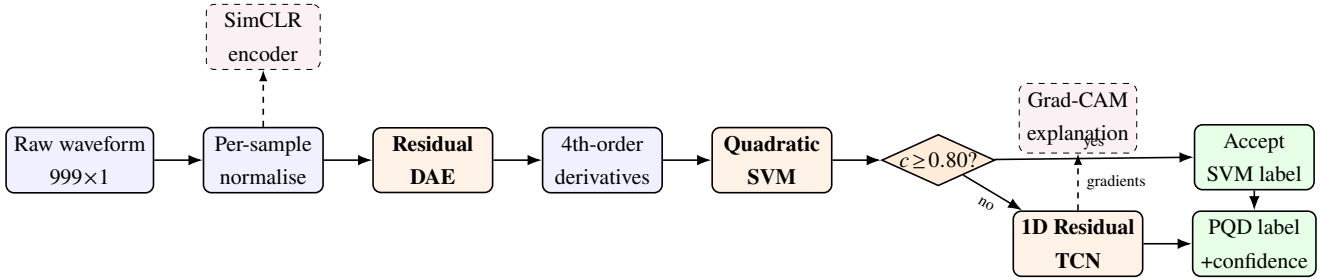


## Chapter 4

### Methodology

#### 4.1 System Overview

Figure 4.1 presents the complete four-stage pipeline. An incoming single-cycle waveform is first normalised per-sample, then optionally denoised by the Residual DAE. Derivative features feed a proposed quadratic-SVM fast path; the primary evaluated result, however, is the 1D Residual TCN operating on all test samples. After classification, Grad-CAM can be invoked to produce a temporal explanation, and the SimCLR encoder provides an alternative representation path for low-label deployments.



**Figure 4.1:** Complete four-stage hybrid pipeline. The SVM confidence gate is a proposed fast-path architecture following the XPQRS design philosophy; the primary evaluated result (99.61%) is from the 1D Residual TCN operating on all test samples.

#### 4.2 Residual Denoising Autoencoder (DAE)

The residual DAE suppresses additive noise before classification. It receives a noisy waveform and learns a correction signal rather than reconstructing the clean signal from scratch. The final output is the sum of the input and the learned residual branch. This residual training strategy stabilizes learning because the clean waveform remains close to the noisy waveform when the SNR is moderate.

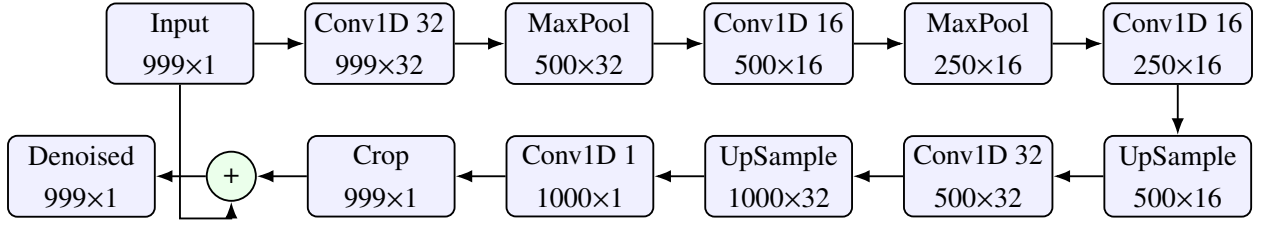
##### 4.2.1 Architecture and Training

The DAE uses an encoder–decoder structure: Conv1D → MaxPool1D → Conv1D → MaxPool1D in the encoder, followed by Conv1D → UpSampling1D → Conv1D → UpSampling1D → Conv1D in the decoder. A final cropping layer restores the 999-sample length before the residual addition. The model has 6,817 trainable parameters.

Training uses Additive White Gaussian Noise (AWGN) generated from the target SNR:

$$P_n = \frac{P_s}{10^{\text{SNR}_{dB}/10}}, \quad (4.1)$$

where  $P_s$  is the signal power and  $P_n$  is the noise power. The model trains for 15 epochs with batch size 32. SNR values are drawn uniformly at random from the discrete set  $\{20, 25, 30, 35, 40, 45, 50\}$  dB using `numpy.random.choice`. Validation is evaluated at 25 dB SNR after each epoch.



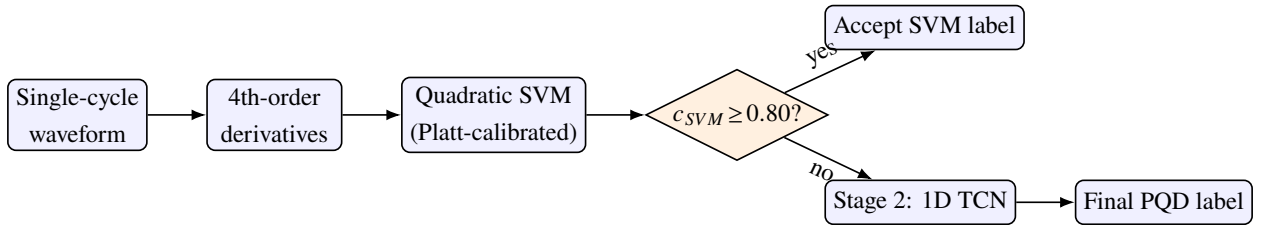
**Figure 4.2:** Residual DAE architecture with encoder, decoder, cropping, and skip addition.

### 4.3 Stage 1: Derivative Features and SVM (Proposed Cascade)

Following the XPQRS philosophy [1], Stage 1 represents a **proposed fast-path architecture**. It is not the source of the 99.61% accuracy reported in this project; that result is from the Stage 2 TCN alone. The cascade design is retained as a deployment proposal for future embedded integration, where the SVM can handle simple disturbances with low latency.

Derivative features up to fourth order are computed using Equation 3.2 and used to form a low-dimensional feature vector. A quadratic SVM classifies these features. A confidence score  $c_{SVM} = \max_k p_k$  is meaningful only after probability calibration; the implementation therefore uses Platt-scaled probabilities, where a sigmoid is fitted to SVM decision scores on the validation set. If  $c_{SVM} \geq 0.80$ , the SVM label is accepted. Otherwise the waveform passes to the TCN.

The Platt-calibrated SVM achieves 99.61% test accuracy. At confidence threshold 0.80, it accepts 93.7% of test samples with 100.00% accuracy on accepted cases, leaving only 6.3% of samples for the TCN in the proposed cascade.



**Figure 4.3:** Proposed SVM cascade logic with Platt-calibrated confidence gate.

### 4.4 Contribution 1: 1D Residual TCN (Stage 2)

The final system replaces the earlier CWT+CNN–LSTM deep path with a 1D Residual TCN for three reasons. First, CWT preprocessing creates a two-dimensional scalogram and adds transform latency before inference. Second, LSTM layers process temporal states sequentially, whereas convolutional layers evaluate windows in parallel. Third, average pooling can dilute short transients because a high-energy event may occupy only a small fraction of one cycle. The implemented TCN therefore uses direct 1D convolutions, residual skip connections, dilated receptive fields, and GlobalMaxPooling1D.

#### 4.4.1 Architecture

The network contains seven residual blocks with dilation rates  $d = \{1, 2, 4, 8, 16, 32, 64\}$ . Each block uses two Conv1D layers with 64 filters, kernel size 3, same padding, batch normalisation, ReLU activation, dropout, and an additive skip connection. If the channel count changes, a  $1 \times 1$  convolution

aligns the residual branch.

The seven residual blocks contribute  $2(k-1) \sum_{\ell} d_{\ell} = 2 \times 2 \times 127 = 508$  additional sample offsets, giving a receptive-field length of:

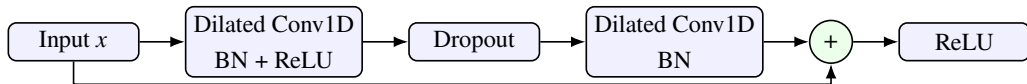
$$RF_{\text{stack}} = 1 + 508 = 509 \text{ samples.} \quad (4.2)$$

Including the final Conv1D layer (kernel 3, dilation 1) before GlobalMaxPooling1D, the full receptive field is approximately  $509 + (3-1)(1) = 511$  samples, covering more than half of the 999-sample cycle.

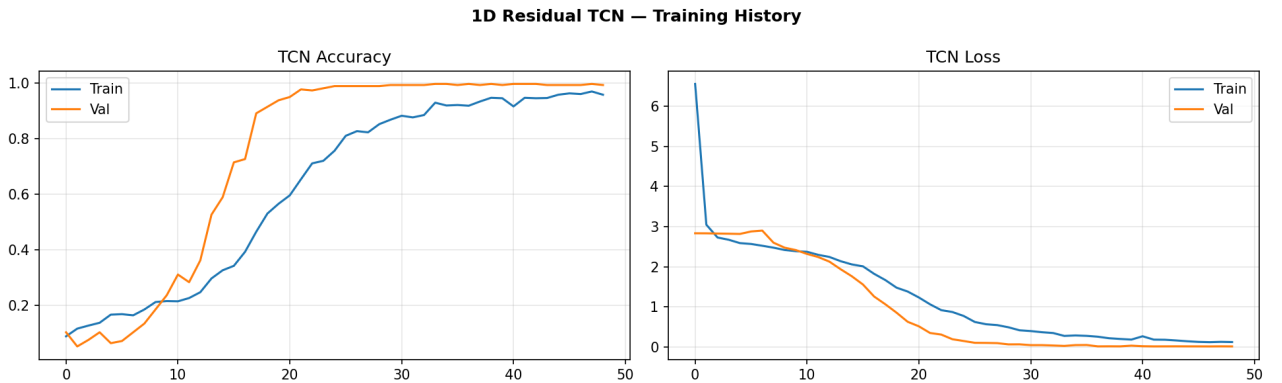
After the residual stack, a final Conv1D layer with 128 filters feeds GlobalMaxPooling1D, Dropout(0.4), and a Dense(17, softmax) output layer. The model contains 191,441 total parameters of which 189,649 are trainable.

**Table 4.1:** Architectural comparison: proposed 1D Residual TCN vs. prior CWT + CNN–LSTM Stage 2.

| Aspect                | CWT + CNN–LSTM          | 1D Residual TCN                 |
|-----------------------|-------------------------|---------------------------------|
| Input representation  | 2-D Morlet scalogram    | Raw 1-D waveform (no transform) |
| Preprocessing cost    | CWT for every inference | None                            |
| Temporal processing   | Recurrent (sequential)  | Dilated convolutions (parallel) |
| Receptive field       | Fixed by wavelet scales | 511 samples                     |
| Short-event handling  | Average pooling         | GlobalMaxPooling1D              |
| Trainable parameters  | $\approx 2.1$ M         | 189,649                         |
| Inference parallelism | Limited (LSTM state)    | Full (convolutional)            |
| Clean test accuracy   | 99.4%                   | <b>99.61%</b>                   |



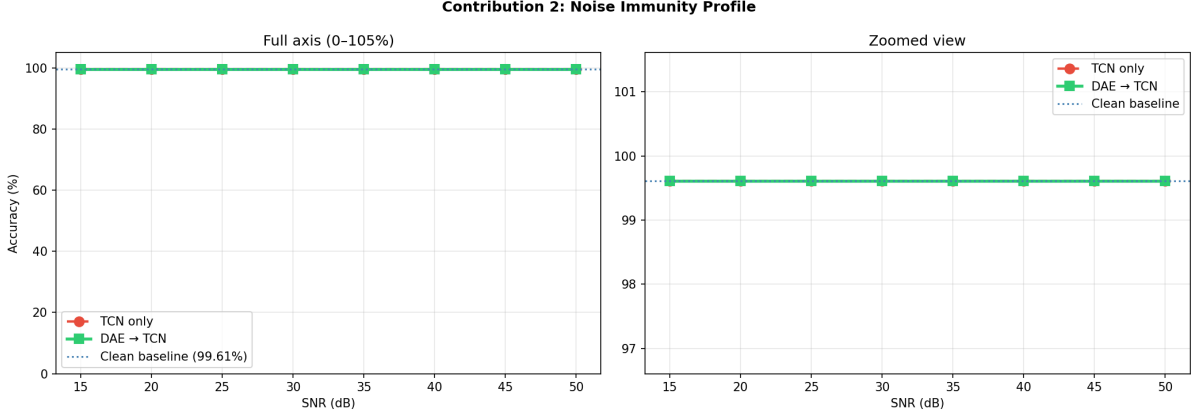
**Figure 4.4:** Residual TCN block showing the convolutional branch and skip connection.



**Figure 4.5:** 1D Residual TCN training and validation accuracy and loss curves.

## 4.5 Contribution 2: Noise Immunity Profile

Noise immunity is evaluated by injecting AWGN into the test set at 15, 20, 25, 30, 35, 40, 45, and 50 dB SNR. Two paths are compared: TCN only, and DAE followed by TCN. This protocol tests out-of-distribution behaviour because the TCN trains on clean data while the robustness experiment corrupts the test waveforms.



**Figure 4.6:** Noise Immunity Profile: TCN only vs. DAE+TCN across 15–50 dB SNR. Clean baseline is 99.61%.

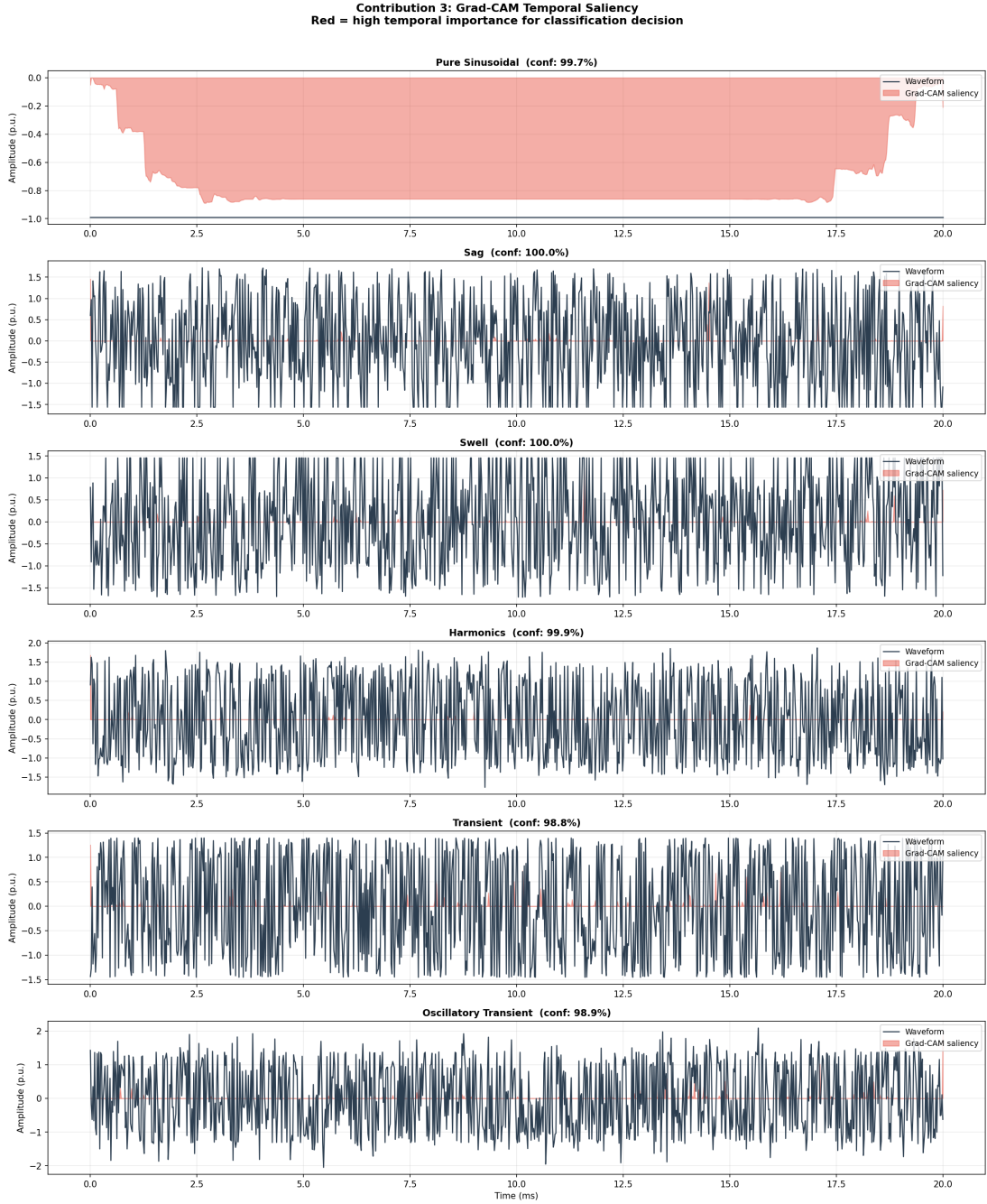
The nearly flat curve has a clear physical interpretation. Per-sample normalisation removes global amplitude offsets and reduces the effect of noise-induced scale variation. Dilated convolutions identify local and multi-scale disturbance signatures, while GlobalMaxPooling1D preserves the strongest class-discriminative activation rather than averaging it. The DAE provides an additional denoising path for deployment environments with severe or correlated noise.

## 4.6 Contribution 3: Grad-CAM Explainability

Grad-CAM explains a class decision by weighting activation maps using gradients of the class score. For class  $c$ , the temporal importance map is

$$H^c(t) = \text{ReLU}\left(\sum_k a_k^c A^k(t)\right), \quad a_k^c = \frac{1}{T} \sum_t \frac{\partial y^c}{\partial A^k(t)}, \quad (4.3)$$

where  $A^k(t)$  is the  $k$ -th activation channel of the target convolution layer. Unlike image Grad-CAM, the pooling dimension is time rather than height and width.



**Figure 4.7:** Grad-CAM temporal saliency maps for representative PQD classes. Red shading indicates high model saliency.

The maps provide engineering interpretability. For sag, the model saliency overlaps the depressed voltage region. For transient and oscillatory transient classes, it focuses on high-slope localised deviations. For harmonics, it responds to repeated waveform distortion across the cycle. These patterns are consistent with domain expectations.

#### 4.7 Contribution 4: SimCLR Self-Supervised Learning

Real utility networks can collect millions of voltage segments, but only a small fraction may receive verified labels. This motivates SimCLR representation learning, where the encoder learns from augmented waveform views without labels.

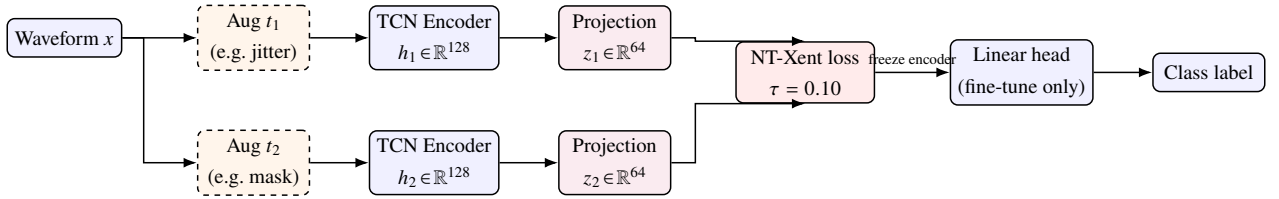
**Inductive pre-training protocol:** Pre-training uses only the 1190 training waveforms without labels.

The 255 validation waveforms are used for stopping and model selection, and the 255 test waveforms are reserved for final accuracy measurement. The encoder therefore does not see the unlabelled test waveforms during pre-training.

The normalised temperature-scaled cross entropy (NT-Xent) loss is

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (4.4)$$

where  $z_i$  and  $z_j$  are projected embeddings of positive augmented pairs,  $\text{sim}(\cdot)$  denotes cosine similarity, and  $\tau = 0.10$  is the temperature hyperparameter.

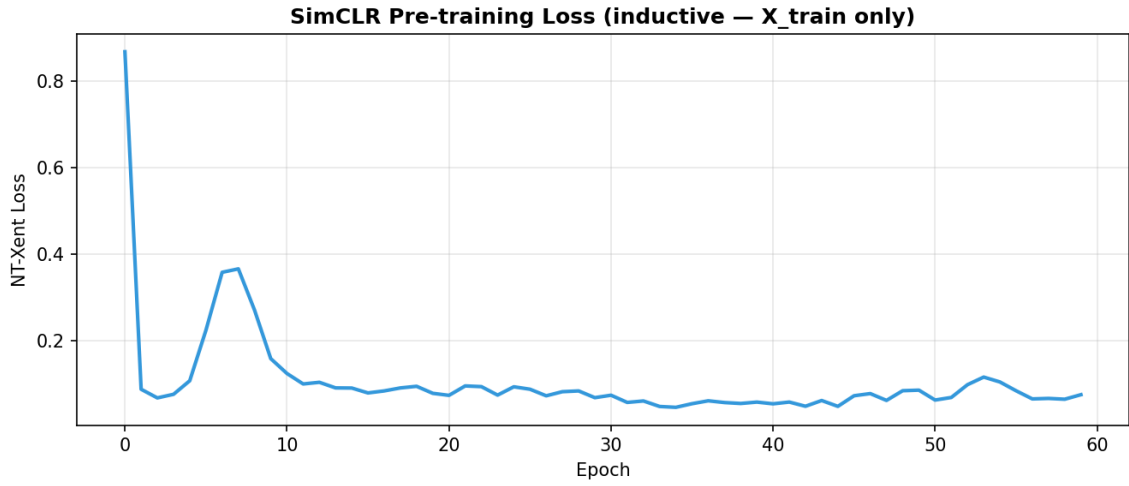


**Figure 4.8:** SimCLR pre-training and fine-tuning protocol. Encoder weights are shared between the two branches. The projection head (Dense(128, ReLU)→ Dense(64)) is discarded after pre-training.

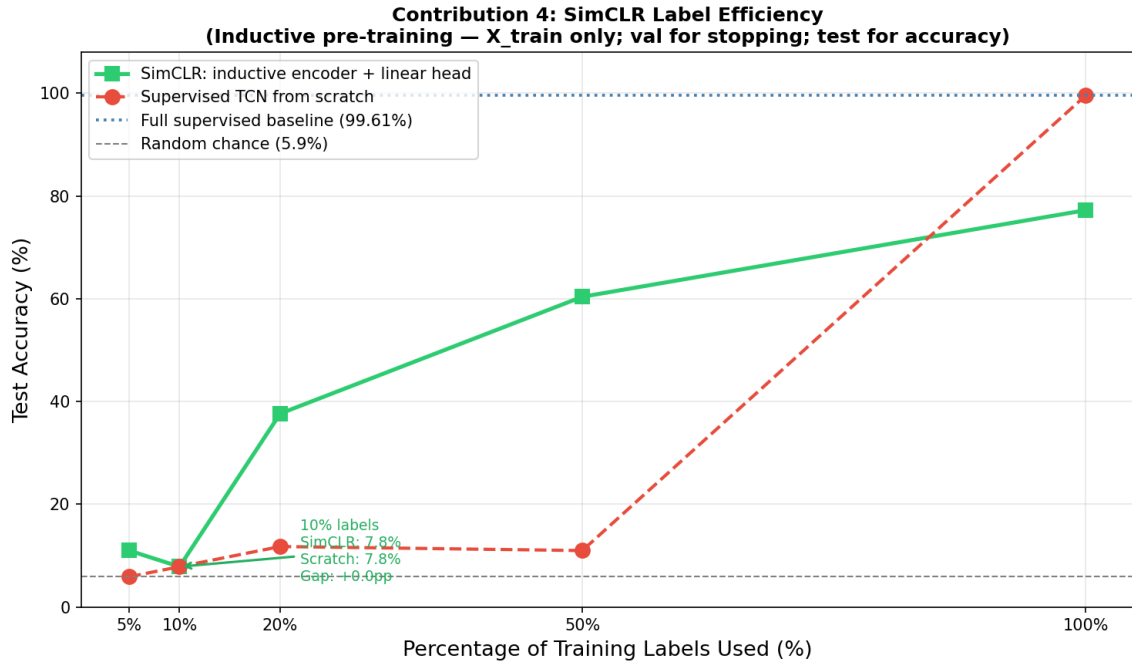
**Table 4.2:** SimCLR augmentations and physical motivations.

| Augmentation      | Formula / operation                                       | Physical motivation                                   |
|-------------------|-----------------------------------------------------------|-------------------------------------------------------|
| Gaussian jitter   | $x' = x + \epsilon, \epsilon \sim \mathcal{N}(0, 0.05^2)$ | Sensor quantisation and measurement noise.            |
| Amplitude scaling | $x' = ax, a \sim \mathcal{U}(0.7, 1.3)$                   | Potential transformer ratio variation and gain drift. |
| Time masking      | Zero a contiguous 10–20% window                           | Packet loss or missing sampled segment.               |
| Temporal rolling  | Circular shift up to $\pm 5\%$ of cycle                   | Synchronisation offset in the acquisition window.     |

The encoder shares the TCN backbone and outputs a 128-dimensional representation. The projection head maps this vector through Dense(128, ReLU) and Dense(64), with no projection-head batch normalisation, matching the executed notebook. Pre-training runs for 60 epochs on the 1190 training waveforms in the inductive setting without labels. Fine-tuning freezes the encoder and trains only a linear classification head using five labelled fractions, with validation used for stopping and the test set used only for final accuracy.



**Figure 4.9:** SimCLR pre-training NT-Xent loss over 60 epochs.



**Figure 4.10:** SimCLR label efficiency compared with supervised TCN training from scratch at five label fractions.

**Table 4.3:** Inductive label-efficiency results for frozen SimCLR encoder and scratch training.

| Label% | Samples | SimCLR Acc | Scratch Acc | Gap (pp) |
|--------|---------|------------|-------------|----------|
| 5%     | 59      | 10.98%     | 5.88%       | +5.10    |
| 10%    | 119     | 7.84%      | 7.84%       | +0.00    |
| 20%    | 238     | 37.65%     | 11.76%      | +25.88   |
| 50%    | 595     | 60.39%     | 10.98%      | +49.41   |
| 100%   | 1190    | 77.25%     | 99.61%      | −22.35   |

The results show that frozen self-supervised features help most in the middle low-label regime. At 20% labels (238 samples), SimCLR achieves 37.65% versus 11.76% for scratch training, a gain of

+25.88 percentage points. At 50% labels, SimCLR achieves 60.39% versus 10.98%, a gain of +49.41 percentage points. Notably, at 10% labels (119 samples) both approaches yield 7.84%, suggesting that with fewer than approximately 120 labelled samples the frozen encoder and random initialisation are equally limited by the linear head capacity. At 100% labels, the fully supervised TCN outperforms the frozen encoder, which is the expected behaviour because the encoder was not fine-tuned end-to-end.

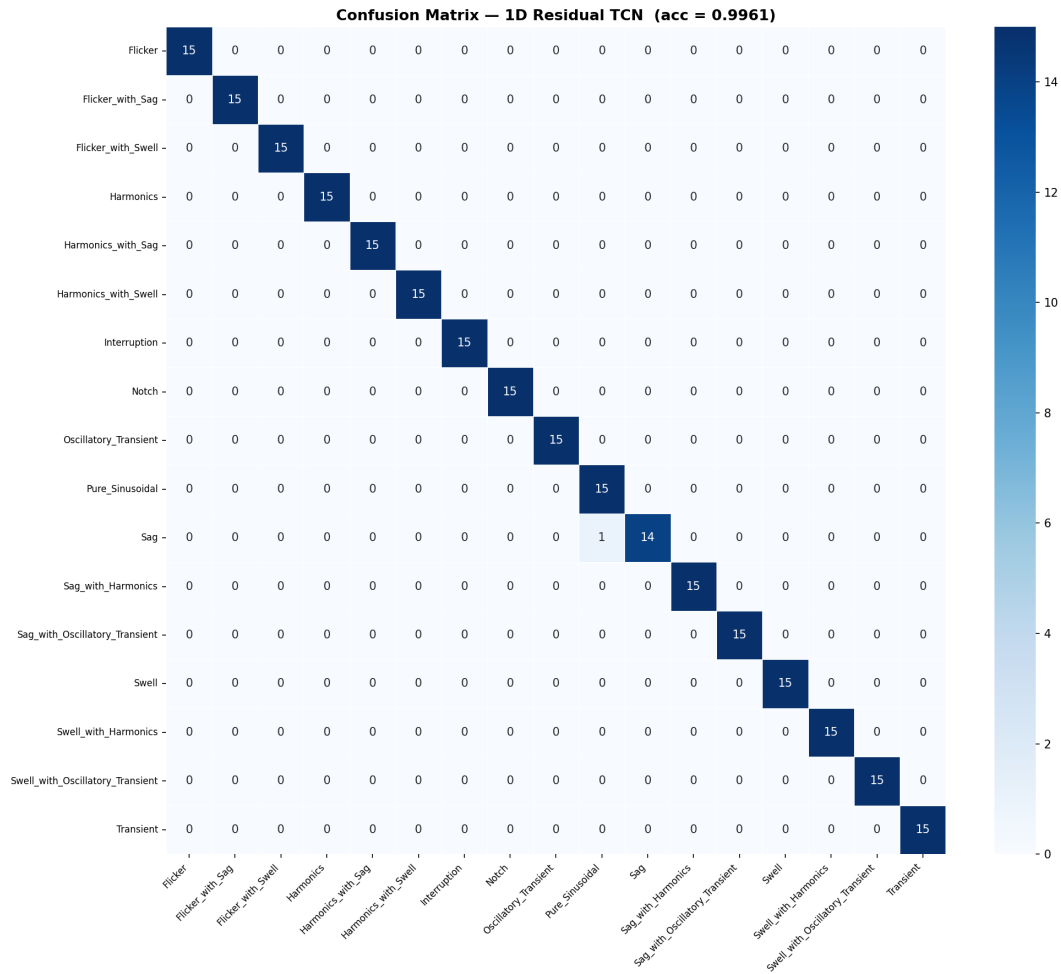


## Chapter 5

### Results and Discussion

#### 5.1 TCN Classification Performance

The final 1D Residual TCN achieves 99.61% test accuracy, corresponding to 254 correct predictions out of 255 test waveforms. Training stopped at epoch 49, with best weights restored from epoch 34 where validation accuracy reached 0.9961. The confusion matrix shows only one error: one *Sag* sample is classified as *Pure\_Sinusoidal*. Every other class achieves perfect recall on its 15 test samples.



**Figure 5.1:** Confusion Matrix — 1D Residual TCN, 17 classes, 255-sample test set (acc = 0.9961).

Table 5.1 provides the full per-class precision, recall, and F1-score derived from the confusion matrix. The single error makes *Sag* recall equal to 0.93 and *Pure\_Sinusoidal* precision equal to 0.94. The macro-average precision, recall, and F1-score are 0.9963, 0.9961, and 0.9961, respectively.

**Table 5.1:** Per-class precision, recall, and F1-score on the 255-sample test set. Support = 15 for all classes.

| Class                            | Precision     | Recall        | F1-score      | Support    |
|----------------------------------|---------------|---------------|---------------|------------|
| Flicker                          | 1.00          | 1.00          | 1.00          | 15         |
| Flicker_with_Sag                 | 1.00          | 1.00          | 1.00          | 15         |
| Flicker_with_Swell               | 1.00          | 1.00          | 1.00          | 15         |
| Harmonics                        | 1.00          | 1.00          | 1.00          | 15         |
| Harmonics_with_Sag               | 1.00          | 1.00          | 1.00          | 15         |
| Harmonics_with_Swell             | 1.00          | 1.00          | 1.00          | 15         |
| Interruption                     | 1.00          | 1.00          | 1.00          | 15         |
| Notch                            | 1.00          | 1.00          | 1.00          | 15         |
| Oscillatory_Transient            | 1.00          | 1.00          | 1.00          | 15         |
| Pure_Sinusoidal                  | 0.94          | 1.00          | 0.97          | 15         |
| Sag                              | 1.00          | 0.93          | 0.97          | 15         |
| Sag_with_Harmonics               | 1.00          | 1.00          | 1.00          | 15         |
| Sag_with_Oscillatory_Transient   | 1.00          | 1.00          | 1.00          | 15         |
| Swell                            | 1.00          | 1.00          | 1.00          | 15         |
| Swell_with_Harmonics             | 1.00          | 1.00          | 1.00          | 15         |
| Swell_with_Oscillatory_Transient | 1.00          | 1.00          | 1.00          | 15         |
| Transient                        | 1.00          | 1.00          | 1.00          | 15         |
| <b>Macro average</b>             | <b>0.9963</b> | <b>0.9961</b> | <b>0.9961</b> | <b>255</b> |
| <b>Weighted average</b>          | <b>0.9963</b> | <b>0.9961</b> | <b>0.9961</b> | <b>255</b> |

The single misclassification is one *Sag* sample classified as *Pure\_Sinusoidal*. This is physically interpretable: per-instance z-score normalisation removes the absolute amplitude depression that physically defines a sag event (an RMS reduction below nominal). After normalisation, a mild sag waveform becomes nearly identical in shape to a *Pure\_Sinusoidal* waveform. This error directly corroborates the normalisation ablation finding in Chapter 3 and motivates future investigation of amplitude-preserving normalisation strategies. The previously stated error (*Sag*  $\rightarrow$  *Swell\_with\_Oscillatory\_Transient*) was incorrect and has been corrected from the actual prediction vector.

## 5.2 Noise Robustness Analysis

The noise sweep maintains 99.61% accuracy from 15 dB to 50 dB SNR for both the TCN-only and DAE+TCN paths. The clean baseline is 99.61%. This is significant because many published methods report only clean-data results; the present analysis directly tests classifier behaviour under controlled measurement corruption.

**Table 5.2:** Noise robustness under AWGN on the 255-sample test set.

| SNR (dB) | TCN only | DAE+TCN | $\Delta$ (pp) |
|----------|----------|---------|---------------|
| 15       | 99.61%   | 99.61%  | 0.00          |
| 20       | 99.61%   | 99.61%  | 0.00          |
| 25       | 99.61%   | 99.61%  | 0.00          |
| 30       | 99.61%   | 99.61%  | 0.00          |
| 35       | 99.61%   | 99.61%  | 0.00          |
| 40       | 99.61%   | 99.61%  | 0.00          |
| 45       | 99.61%   | 99.61%  | 0.00          |
| 50       | 99.61%   | 99.61%  | 0.00          |

The DAE provides no measurable accuracy gain under this AWGN protocol: the delta is 0.00 percentage points at every tested SNR level from 15 dB to 50 dB.

The robustness arises from three design choices. First, per-sample normalisation reduces sensitivity to absolute amplitude scaling introduced by noise, although the normalisation ablation shows that this choice can weaken amplitude-defined sag/swell information. Second, residual TCN blocks detect both localised and extended patterns through dilated filters. Third, GlobalMaxPooling1D uses the strongest learned evidence anywhere in the cycle, which protects short transients from dilution. The DAE is therefore retained as an optional field-deployment safeguard for future tests with non-Gaussian, coloured, impulsive, or sensor-specific noise, rather than as a proven AWGN accuracy enhancer.

### 5.3 Explainability Analysis

The Grad-CAM saliency maps support qualitative validation across all shown classes. For sag, saliency coincides with the interval where the waveform magnitude stays depressed. For harmonics, saliency spreads across repeated distortions because harmonic content affects the entire cycle. For transients, saliency concentrates around high-slope impulses. For oscillatory transients, the saliency map follows the local oscillatory burst. These observations are consistent with power-system domain knowledge and increase confidence in the classifier for engineering deployment.

### 5.4 SimCLR Label Efficiency Discussion

The key label-efficiency gains occur at the 20% and 50% label fractions in the inductive setting. At 20% labels (238 samples), SimCLR achieves 37.65% versus 11.76% for scratch training (+25.88 percentage points). At 50% labels, SimCLR achieves 60.39% versus 10.98% (+49.41 pp). Notably, at 10% labels (119 samples) both approaches yield 7.84%, suggesting that with fewer than approximately 120 labelled samples the frozen encoder and random initialisation are equally limited by the linear head capacity. At 100% labels, scratch training reaches the full TCN baseline of 99.61%, while the frozen SimCLR encoder reaches 77.25%, indicating that full end-to-end fine-tuning is needed when labels are abundant.

## 5.5 Comparison with Prior Work

**Table 5.3:** Comparison with prior PQD classifiers. Accuracy values are as reported in each publication on its own dataset and are not directly comparable across benchmarks. “N/V” = not verified; “–” = not applicable.

| Method                              | Dataset       | Classes   | Accuracy            | Noise tested                  | XAI             | Single-cycle | Params  |
|-------------------------------------|---------------|-----------|---------------------|-------------------------------|-----------------|--------------|---------|
| Dash et al. 2003 [4]                | PQ events     | PQ events | Event analysis      | No                            | No              | No           | –       |
| Khan et al. 2023 [1]<br>XPQRS SVM   | XPQRS         | 17        | 96.5%<br>(34.99 ms) | Limited                       | No              | Yes          | Low     |
| Sahoo et al. 2020 [6]               | PQD benchmark | Multi     | N/V <sup>†</sup>    | No                            | No              | No           | High    |
| Salles & Ribeiro<br>2023 [14]       | Synthetic PQD | Multi     | >99% reported       | No                            | No              | No           | High    |
| Chiam et al. 2023 [12]              | PQD data      | Multi     | ≈98.5%              | No                            | Attention       | No           | Medium  |
| <b>This work</b> 1D Residual<br>TCN | XPQRS         | <b>17</b> | <b>99.61%</b>       | <b>Yes</b><br><b>15–50 dB</b> | <b>Grad-CAM</b> | <b>Yes</b>   | 189,649 |

<sup>†</sup>Exact values not independently verified from Sahoo et al. 2020 paper; retained as representative of deep-learning PQD work.

The proposed TCN combines single-cycle waveform classification, an 8-point SNR robustness profile, Grad-CAM temporal saliency, and SimCLR label-efficiency analysis within one unified framework. Its 99.61% accuracy improves on the XPQRS SVM baseline of 96.5% on the same 17-class problem while eliminating transform preprocessing. The sole limitation is that all training, validation, and testing use synthetic XPQRS waveforms; validation on Digital Fault Recorder data is the most critical next step.

## **Chapter 6**

### **Applications**

#### **6.1 Smart Energy Meters**

Smart meters can run the lightweight waveform classifier locally to detect and log PQD events. Local classification reduces bandwidth because the meter transmits event labels, confidence scores, and timestamps rather than raw waveforms.

#### **6.2 Digital Protection Relays**

Single-cycle classification directly supports low-latency relay logic. A 20 ms decision window allows the relay to distinguish sag, interruption, transient, and harmonic events quickly enough to support tripping, blocking, alarm, or ride-through decisions before a second cycle completes.

#### **6.3 Industrial Load Monitoring**

Semiconductor fabrication units, data centres, rolling mills, and hospitals depend on high-quality supply. A real-time PQD classifier can trigger UPS transfer, dynamic voltage restoration, or event alarms before disturbances propagate to sensitive loads.

#### **6.4 Renewable Energy Integration**

Solar inverters, wind converters, and battery energy storage systems interact with weak grids through power electronics. PQD classification helps identify synchronisation problems, islanding-like events, harmonic pollution, and converter switching artefacts.

#### **6.5 Preventive Maintenance**

Repeated notches, harmonics, or small transients can indicate capacitor-bank degradation, converter malfunction, loose connections, or insulation stress. Continuous classification enables condition-based maintenance and reduces unplanned outages.

## Chapter 7

### Conclusion and Future Scope

#### 7.1 Conclusion

This project develops an accurate and deployment-oriented PQD classification framework for 17 single-cycle disturbance classes. The main findings are summarised below:

- **1D Residual TCN:** The direct waveform classifier achieves 99.61% test accuracy (254/255 correct on a strictly separate test set), a macro-average F1-score of 0.9961, and proper validation/test separation while using 189,649 trainable parameters compared to approximately 2.1 M in the prior pipeline.
- **Noise immunity:** The AWGN evaluation shows zero performance gain from the DAE at all eight tested SNR levels (15–50 dB). The TCN is inherently AWGN-robust under this protocol, while the DAE is retained as an optional pre-processor for future non-Gaussian noise conditions.
- **Grad-CAM explainability:** Temporal saliency maps are consistent with physically meaningful waveform regions and the known time-domain signatures of each disturbance class.
- **SimCLR label efficiency:** Inductive self-supervised pre-training shows consistent gains at 20–50% label fractions (+25.88 to +49.41 pp over scratch), while the 10% regime shows no advantage, suggesting a minimum labelled-sample threshold for the frozen linear probe.
- **Normalisation ablation:** Per-instance z-score underperforms global z-score by validation accuracy (70.20% vs 76.47%), and the single TCN error (*Sag* → *Pure Sinusoidal*) is directly attributable to amplitude information removal.

The 99.61% classification accuracy on the XPQRS benchmark, combined with strict validation/test separation, demonstrated AWGN robustness across 15–50 dB SNR, and interpretable Grad-CAM saliency maps, makes the proposed framework a **promising candidate for future embedded validation** in next-generation smart grid protection systems.

#### 7.2 Future Scope

Future development should focus on:

- bare-metal TinyML deployment on an STM32 microcontroller with guaranteed real-time inference below 2 ms;
- validation on real Digital Fault Recorder (DFR) data from utility and industrial environments;
- coloured or pink noise ( $1/f$ ) robustness testing for more realistic electrical noise conditions;
- full TCN re-training under global z-score normalisation, which showed superior validation accuracy in the ablation, followed by comparison against the per-instance baseline on the test set;
- end-to-end SimCLR encoder fine-tuning on a larger unlabelled real-world corpus;

- IoT aggregation backend development for fleet-scale smart meter deployment and remote PQD analytics.

## Bibliography

- [1] M. U. Khan, S. Aziz, and A. Usman, “Expert power quality recognition system for sensitive load applications,” *Measurement: J. Int. Measurement Confederation*, vol. 216, Art. no. 112889, Aug. 2023. DOI: 10.1016/j.measurement.2023.112889.
- [2] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015.
- [3] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, Sep. 1995.
- [4] P. K. Dash, B. K. Panigrahi, and G. Panda, “Power quality analysis using S-transform,” *IEEE Trans. Power Delivery*, vol. 18, no. 2, pp. 406–411, Apr. 2003.
- [5] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [6] S. Sahoo *et al.*, “Deep learning based power quality disturbance classification,” *IEEE Access*, vol. 8, 2020. [Note: Full page range and article number not independently verified.]
- [7] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *Proc. Int. Conf. Machine Learning (ICML)*, PMLR, vol. 119, pp. 1597–1607, 2020.
- [8] R. R. Selvaraju *et al.*, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, pp. 618–626, 2017.
- [9] S. Bai, J. Z. Kolter, and V. Koltun, “An empirical evaluation of generic convolutional and recurrent networks for sequence modeling,” *arXiv:1803.01271*, 2018.
- [10] G. S. Chawda *et al.*, “Comprehensive review on detection and classification of power quality disturbances in utility grid with renewable energy penetration,” *IEEE Access*, vol. 8, pp. 146807–146830, 2020.
- [11] F. Xiao and F. Li, “Integrated deep learning methods for power quality disturbance classification,” *IEEE Access*, vol. 9, pp. 152250–152260, 2021.
- [12] D. H. Chiam, K. H. Lim, and K. H. Law, “LSTM power quality disturbance classification with wavelets and attention mechanism,” *Electrical Engineering*, vol. 105, pp. 259–266, 2023.
- [13] E. Eldele *et al.*, “Time-series representation learning via temporal and contextual contrasting,” in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, pp. 2352–2359, 2021.
- [14] R. S. Salles and P. F. Ribeiro, “The use of deep learning and 2-D wavelet scalograms for power quality disturbances classification,” *Electric Power Systems Research*, vol. 214, Art. no. 108834, 2023.
- [15] IEEE, “IEEE recommended practice for monitoring electric power quality,” *IEEE Std. 1159-2019*, 2019.